03 Assignment3 Soap Integration

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Assignment 3 — External SOAP Partner (mapped to ShipTrack, live not paper)

TeamID: TODO | Name/Roll: TODO

CampusEats asked for integration.pdf, partner.wsdl, soap-request/response/fault.xml (design-time only). ShipTrack implements the same edge live.

1. Context — why SOAP here while rest is REST

Edge: pricing-service → location-service CountryInfo (logistics reference-data edge). 3-sentence justification: strong WSDL contract both sides can codegen; message-level GetCountryInfo lookup must stay stable while REST CRUD evolves; fits currency/country reference data shared like an external partner. Rest stays REST for CRUD speed. See docs/architecture.md:71-72.

2. Partner WSDL (all six elements)

Live files (not standalone partner.wsdl):
- services/location-service/src/soap/countryWsdl.js:1-59 — types (GetCountryInfoRequest/Response), message In/Out, portType CountryInfoPortType, binding SOAP document/literal over http://schemas.xmlsoap.org/soap/http, service CountryInfoService → http://localhost:8007/wsdl/country.
- services/pricing-service/src/soap/currencyWsdl.js — same shape for currency conversion.
- Served by node-soap (src/soap/index.js per service).
- Prove: curl -s http://localhost:8007/wsdl/country?wsdl | head -60 → 200 XML; currency at http://localhost:8006/wsdl/currency?wsdl.

3. SOAP envelopes (request / response / fault)

No standalone soap-*.xml — live equivalents:
- Call: client.GetCountryInfoAsync({ countryCode }) in services/pricing-service/src/soap/countryClient.js:37-42.
- Request = Header (credentials) + Body (GetCountryInfo); success = {code, name, capital, currency}; fault (unknown/unavailable) → null / SOAP Fault.
- Tests: pricing-service/tests/currencySoap.test.js, countryClient.test.js, location-service/tests/countrySoap.test.js.
- Run: npm test --workspace services/pricing-service (DB mocked, no Docker needed).

4. HTTP binding

POST /wsdl/country, Host localhost:8007, Content-Type: text/xml; charset=utf-8, SOAPAction: "http://shiptrack.local/countryinfo/GetCountryInfo" (countryWsdl.js:48), endpoint from soap:address (countryWsdl.js:56). Caller overrides baked localhost via setEndpoint(${LOCATION_SERVICE_URL}/wsdl/country) (countryClient.js:12-22).

```
curl -i -X POST http://localhost:8007/wsdl/country \
  -H 'Content-Type: text/xml' \
  -H 'SOAPAction: "http://shiptrack.local/countryinfo/GetCountryInfo"' \
  --data-binary @soap-request.xml
```

5. Discovery (modern form, no UDDI server)

Via env + WSDL URL: LOCATION_SERVICE_URL (pricing-service/.env.example, countryClient.js:12) + fetch ${LOCATION_SERVICE_URL}/wsdl/country?wsdl. Lazy soap.createClientAsync(wsdlUrl) with cache-clear on fail (countryClient.js:10-30). Registry row: business=ShipTrack-Location, service=CountryInfoService, endpoint=http://location-service:8007/wsdl/country, tModel=http://shiptrack.local/countryinfo + ?wsdl.

6. Fault mapping (partner vocab must not leak)

countryClient.js:40 returns null on unknown/SOAP-fail → pricingService.js degrades to default transit/currency; shipment-service/src/clients/trackingClient.js:9 fallback: () => null so status update still succeeds. Surfaced only as REST ApiError (common/src/errors.js): 404 NOT_FOUND, 422 UNPROCESSABLE_ENTITY, 503 SERVICE_UNAVAILABLE — never raw SOAP fault to students.